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Reserve Bank of India (Commercial Banks – Credit Valuation Adjustment Framework) Directions, 2026

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Introduction

In exercise of the powers conferred by Section 35A of the Banking Regulation Act, 1949, and all other provisions / laws enabling the Reserve Bank of India ('RBI') in this regard, the RBI being satisfied that it is necessary and expedient in the public interest so to do, hereby issues the Directions hereinafter specified.

Chapter-I: Preliminary

A. Short Title and Commencement

1. These Directions shall be called the Reserve Bank of India (Commercial Banks – Credit Valuation Adjustment Framework) Directions, 2026.

2. These Directions shall come into effect from April 1, 2027.

B. Applicability

3. These Directions shall be applicable to Commercial Banks (hereinafter collectively referred to as 'banks' and individually as a 'bank').

   For the purpose of these Directions, 'Commercial Banks' means banking companies (other than Small Finance Banks, Payments Banks, and Local Area Banks), corresponding new banks, and the State Bank of India, as defined respectively under clauses (c), (da), and (nc) of Section 5 of the Banking Regulation Act, 1949.

C. Definitions

4. In these Directions, unless the context states otherwise, the terms herein shall bear the meanings assigned to them below:

   (1) **'Banking Book'** shall have the same meaning as assigned to it in paragraph 4(1) of the Reserve Bank of India (Commercial Banks - Prudential Norms on Capital Adequacy) Directions, 2025.

   (2) **'Covered Transactions'** include all derivatives except those transacted directly with a qualified central counterparty (QCCP) and except those transactions meeting the conditions set out at paragraphs 85(6)(i)(f) and 85(6)(i)(g) of the Reserve Bank of India (Commercial Banks - Prudential Norms on Capital Adequacy) Directions, 2025 and paragraphs 16 (6) to

16(8) of the Reserve Bank of India (Commercial Banks - Forthcoming Instructions) Amendment Directions, 2026.

(3) **'Credit Valuation Adjustment' (CVA)**, is specified at a counterparty level, and reflects the adjustment to default risk-free price of a derivative or Security Financing Transaction (SFT) due to a potential default of the counterparty. It is an adjustment to the valuation of a derivative transaction or SFT to account for the credit risk of contracting parties. In the context of these Directions, it means regulatory CVA and may differ from accounting CVA, on account of the following:

(a) regulatory CVA excludes the effect of the bank's own default; and

(b) constraints arising from best practices in accounting CVA which are imposed in computation of regulatory CVA.

(4) **'CVA risk'** is the risk of losses arising from change in CVA values in response to changes in counterparty credit spreads and market risk factors that drive prices of derivative transactions and SFTs.

(5) **'Derivative'** shall have the same meaning as assigned to it in Section 45U(a) of the RBI Act, 1934.

(6) **'Netting Set'** shall have the same meaning as assigned to it in paragraph 4(23) of the Reserve Bank of India (Commercial Banks - Prudential Norms on Capital Adequacy) Directions, 2025.

(7) **'Qualifying central counterparty' (QCCP)** shall have the same meaning as assigned to it in paragraph 4(29) of the Reserve Bank of India (Commercial Banks - Prudential Norms on Capital Adequacy) Directions, 2025.

(8) **'Securities financing transaction' (SFT)** shall have the same meaning as assigned to it in paragraph 4(30) of the Reserve Bank of India (Commercial Banks - Prudential Norms on Capital Adequacy) Directions, 2025.

(9) **'Trading Book'** shall have the same meaning as assigned to it in paragraph 4(34) of the Reserve Bank of India (Commercial Banks - Prudential Norms on Capital Adequacy) Directions, 2025.

5. All other expressions, unless defined herein, shall have the same meaning as have been assigned to them under the applicable Acts, rules / regulations made thereunder, or any statutory modification or re-enactment thereto, or as used in commercial parlance, as the case may be.

Chapter-II: Scope of Application

6. A bank shall calculate the capital requirements for Credit Valuation Adjustment (CVA) risk on a standalone basis for all covered transactions in both banking book and trading book. It includes CVA for a bank's entire portfolio of covered transactions and eligible CVA hedges. Trades pertaining to SFTs shall be excluded from the calculation of the CVA capital charge.

Explanation: (1) Where a derivative transaction is subsequently novated to a QCCP, the resulting exposure upon novation shall be treated in the same manner as an exposure arising from a transaction undertaken directly with the QCCP and shall, accordingly, be excluded from the scope of Covered Transactions.

(2) A bank that is a clearing member of a CCP shall capitalise its CVA risk exposure to clients as bilateral trades, irrespective of whether the clearing member guarantees the trade or acts as an intermediary between the client and the CCP. However, to recognise the shorter close-out period applicable to cleared transactions, a clearing member may calculate its exposure to clients using an MPOR of at least five business days for the purpose of computing the exposure at default under the SA-CCR framework. The resulting exposure at default shall be used for the calculation of the CVA capital requirement.

(3) For the purposes of this paragraph, "Covered Transactions" shall include all eligible transactions of the bank globally, comprising transactions undertaken through both its domestic operations and overseas branches. Transactions booked through different branches of the bank with the same counterparty may be treated as part of a single netting set for CVA purposes only where the relevant netting agreement independently satisfies the applicable requirements for recognition as a netting agreement, including legal enforceability in all relevant jurisdictions in which such transactions are booked.

Chapter-III: Approaches for CVA Risk Capital Charge

A. Basic Approach (BA-CVA)

7. (1) A bank shall, for computation of capital charge for CVA, use the Basic Approach for Credit Valuation Adjustment (BA-CVA), as provided in Chapter IV

of these Directions, subject to the exception provided in Section B under Chapter III of these Directions.

(2) The detailed illustrations for computation of CVA Framework are given in **Annex I**.

B. Alternate Treatment

8. A bank whose aggregate notional amount of non-centrally cleared derivatives, on a consolidated group-wide basis, is less than or equal to ₹ 10 lakh crore as at the reporting date, may opt not to calculate its CVA capital requirements using the BA-CVA and may instead adopt an alternative treatment.

Provided that, the supervisory authority may not permit this option for a bank, if it determines that CVA risk resulting from the bank's derivative positions materially contributes to the bank's overall risk.

Explanation: Where a bank exceeds the prescribed threshold as at the reporting date, it shall apply the BA-CVA with effect from that reporting date.

9. The capital requirement for CVA risk under the alternate treatment is as follows:

(1) A bank shall set its CVA capital requirement equal to 100 per cent of its capital requirement for counterparty credit risk (CCR), computed as per the Reserve Bank of India (Commercial Banks – Forthcoming Instructions) Amendment Directions, 2026.

(2) A bank is not allowed to recognise CVA hedges.

(3) A bank shall apply this treatment to its entire portfolio of covered transactions.

C. Capital Requirements and Risk Weighted Assets

10. The capital charge for CVA risk shall be calculated as under:

(1) For a bank adopting the alternate treatment, the capital charge shall be calculated as per paragraph 9 of these Directions.

(2) For a bank adopting the reduced version of BA-CVA, the capital charge shall be calculated as per paragraph 13 of these Directions.

(3) For a bank adopting the full version of BA-CVA, the capital charge shall be calculated as per paragraph 20 of these Directions.

11. The risk-weighted assets for CVA risk shall be determined by multiplying the capital charge, calculated as provided in paragraph 10 of these Directions, by 12.5.

Chapter-IV: CVA risk capital charge under the BA-CVA

12. A bank can, at its discretion, choose to implement either the full version or the reduced version of BA-CVA. However, a bank shall calculate the reduced version of BA-CVA capital requirements, as it is also part of the full BA-CVA capital calculations as a conservative means to limit hedging recognition. The full version of BA-CVA recognises counterparty credit spread hedges and is intended for a bank that hedges CVA risk. The reduced version is designed to simplify BA-CVA implementation for a less sophisticated bank that does not hedge CVA risk.

A. Reduced version of the BA-CVA (hedges are not recognized)

13. The capital requirements for CVA risk under the reduced version of the BA-CVA is calculated using the following formula:

$$CVA_{reduced} = DS \times K_{reduced}$$

Where,

DS is the discount scalar = 0.65; and

$K_{reduced}$ is calculated as provided in paragraph 14 of these Directions.

14. A bank shall compute $K_{reduced}$ using the following formula:

$$K_{reduced} = \sqrt{\left(\rho * \sum_{C} SCVA_C\right)^2 + (1 - \rho^2) * \sum_{C} {SCVA_C}^2}$$

Where,

(i) The summations shall cover all counterparties that are within scope of the CVA charge.

(ii) $SCVA_c$ is the CVA capital requirement that counterparty c would receive if considered on a stand-alone basis (referred to as "stand-alone CVA capital"). The calculation of standalone CVA capital shall be as provided in paragraph 15 of these Directions.

(iii) ρ = 50 per cent is the supervisory correlation parameter. ρ^2 = 25 per cent, represents the correlation between credit spreads of any two counterparties. The effect of ρ is to recognise the fact that the CVA risk to which a bank is exposed to is less than the sum of the CVA risk for each counterparty, given that the credit spreads of counterparties are typically not perfectly correlated. One of the basic assumptions underlying the reduced BA-CVA is that systematic credit spread risk is driven by a single factor. Under this assumption, ρ can be interpreted as the correlation between the credit spread of a counterparty and the single credit spread systematic factor.

(iv) The first and second terms under the square root in the formula above aggregates the systematic components and idiosyncratic components of CVA risk respectively.

15. The stand-alone CVA capital requirement for counterparty c ($SCVA_c$) used in the formula in paragraph 14 of these Directions shall be calculated using the following formula:

$$SCVA_C = \frac{1}{\alpha} * RW_C * \sum_{NS} M_{NS} * EAD_{NS} * DF_{NS}$$

Where,

(i) the summation shall be across all netting sets with the counterparty c.

(ii) $\mathrm{RW_C}$ is the risk weight for counterparty c and reflects the volatility of its credit spread. These risk weights are based on a combination of sector and credit quality of the counterparty, as prescribed in paragraph 16 of these Directions.

(iii) $\mathrm{M_{NS}}$ is the effective maturity for the netting set NS, as prescribed in paragraph 17 of these Directions.

(iv) $\mathrm{EAD}_{\mathrm{NS}}$ is the exposure at default (EAD) of the netting set NS, calculated in the same way as the bank calculates it as per the Standardised Approach for Counterparty Credit Risk (SA-CCR) in the Reserve Bank of India (Commercial Banks – Forthcoming Instructions) Amendment Directions, 2026.

(v) (a) $\mathrm{DF}_{\mathrm{NS}}$ is a supervisory discount factor, and is calculated using the following formula:

$$DF_{NS} = \frac{1 - e^{-0.05 * M_{NS}}}{0.05 * M_{NS}}$$

(b) DF is the supervisory discount factor averaged over time between today and the netting set's effective maturity date. The interest rate used for discounting is set at 5 per cent, hence 0.05 in the formula. The product of EAD and effective maturity in the BA-CVA formula is a proxy for the area under the discounted expected exposure profile of the netting set. The netting set's effective maturity is defined as an average of actual trade maturities. This definition lacks discounting and is compensated by adding the supervisory discount factor.

(vi) α set at 1.4 and is the multiplier used to convert Effective Expected Positive Exposure (EEPE) to EAD in the SA-CCR in the Reserve Bank of India (Commercial Banks – Forthcoming Instructions) Amendment Directions, 2026. It is used to convert the EAD of the netting set back to EEPE.

16. The supervisory risk weights (RW_c in computation of stand-alone CVA capital requirement under paragraph 15 of these Directions or RW_h in computation of SNH_C under paragraph 23 of these Directions and for calculation of RW_i in computation of IH under paragraph 24 of these Directions) are provided in **Table 1** below. Credit quality of the counterparty is specified as investment grade (IG), high yield (HY), or not rated (NR) depending on the external ratings or lack of it, as following:

(a) Credit rating below BBB / Baa -shall be considered as HY. For the purpose of mapping external ratings assigned by Eligible Credit Rating Agencies (ECRA), a bank shall refer to Table 20 and Table 22 of the Reserve Bank

of India (Commercial Banks – Capital Charge for Credit Risk – Standardised Approach) Directions, 2026, as amended from time to time, and shall use the rating mapping specified in the respective tables. The supervisory risk weight shall, however, be determined in accordance with Table 1 of these Directions. Where a "+" or "–" notation is attached to a rating, the corresponding risk weight for the main rating category shall apply.

(b) In case a counterparty has more than one credit rating, the worst of the ratings shall be used to decide on the credit quality and resultant risk weight. Where there are three or more ratings with different risk weights, the ratings corresponding to the two lowest risk weights should be referred to and the higher of those two risk weights shall be applied.

(c) Where there are no external ratings, the risk weights corresponding to NR is to be applied.

(d) "A bank shall put in place an internal policy, approved by the appropriate authority, governing the granular classification of sectors, with such classification mapped to the sectors specified in Table 1. The policy shall be applied consistently across all counterparties. Where a counterparty cannot be assigned to any of the sectors specified in Table 1, the exposure to such counterparty shall be assigned to the "Other Sectors" category."

(e) A bank shall use the ratings assigned by Eligible Credit Rating Agencies (ECRA), as specified in paragraph 24 of the Reserve Bank of India (Commercial Banks – Capital Charge for Credit Risk – Standardised Approach) Directions, 2026, as amended from time to time, for the purpose of assigning supervisory risk weights for CVA risk.

(f) For the purpose of determining the credit quality of a counterparty under these Directions, a bank shall use the issuer rating applicable to that counterparty.

Table 1 - Supervisory risk weights

Sr. No.	Sector of Counterparty	Credit Quality of Counterparty	
		IG	HY and NR

1	Sovereigns including central banks and multilateral development banks	0.5%	2.0%
2	Local government, government-backed non-financials, education and public administration	1.0%	4.0%
3	Financials including government-backed financials	5.0%	12.0%
4	Basic materials, energy, industrials, agriculture, manufacturing, mining and quarrying	3.0%	7.0%
5	Consumer goods and services, transportation and storage, administrative and support service activities	3.0%	8.5%
6	Technology, telecommunications	2.0%	5.5%
7	Health care, utilities, professional and technical activities	1.5%	5.0%
8	Other sectors	5.0%	12.0%

17. A bank shall calculate the effective maturity M_{NS} as under:

(1) The effective maturity (M_{NS}) is subject to a floor of one year, subject to the exemption provided in sub-paragraph 17(5) below.

Explanation: A transaction with an original maturity exceeding one year shall remain subject to the one-year maturity floor, notwithstanding that its residual maturity may subsequently fall below one year.

(2) For an instrument subject to a determined cash flow schedule, effective maturity M for the instrument is defined as follows:

$$Effective\ maturity\ for\ an\ instrument = M = \sum_t t * CF_t \Big/ \sum_t CF_t$$

Where,

CF denotes the cash flows (principal, interest payments, and fees) contractually payable by the borrower / counterparty in period t;

Provided that, if the bank is not in a position to calculate the effective maturity of the contracted payments as noted above, it shall use a more conservative measure of M such as that M equals the maximum remaining time (in years) that the borrower / counterparty is permitted to take to fully discharge its contractual obligation (principal, interest, and fees) under the terms of the agreement. Normally, this will correspond to the nominal maturity of the instrument.

(3) For transactions that are not subject to a legally enforceable bilateral netting agreement (referred to hereafter as 'netting agreement') that is recognised

for regulatory capital purposes, each such transaction is interpreted as its own netting set. For such transactions, the effective maturity of the instrument, M = M_{NS}.

(4) For transactions that are part of a netting agreement, the effective maturity of the netting set M_{NS} shall be the weighted average maturity (M) of the transactions within the netting agreement. Further, the notional amount of each transaction shall be used for weighting the maturity.

Explanation (1): For the purpose of calculating M_{NS}, the notional amount of each transaction shall refer to the contractual trade notional amount. Adjusted notional amounts or effective notional amounts used for SA-CCR shall not be used for weighting maturity.

(2) Where a netting agreement comprises transactions that satisfy the collateralisation and documentation conditions specified in sub-paragraph 17(5) as well as transactions that do not, the transactions shall be grouped separately for the purpose of computing the weighted average maturity and applying the applicable maturity floor to each group.

(3) For the purposes of paragraph 17(4), M_{NS} shall first be calculated as the notional-weighted average of the effective maturities of the transactions comprised in the relevant group, prior to application of the applicable maturity floor. The applicable maturity floor under sub-paragraph 17(1) or 17(5), as the case may be, shall thereafter be applied to the resulting weighted average maturity.

(5) The one-year floor for M_{NS}, prescribed in sub-paragraph 17(1) above, shall not apply to fully or nearly-fully collateralised OTC derivatives transactions with an original maturity of less than one year, where the documentation contains daily re-margining clauses. Fully or nearly-fully collateralised OTC derivative transactions shall mean cases where variation margin covers the current mark-to-market exposure resulting from changes in the market value of a derivative contract. The intention is to include both parties of a transaction meeting these conditions where neither of the parties is systematically under-collateralised. To qualify for this exemption, the documentation of such eligible transactions shall require daily revaluation and must include provisions that allow for the prompt liquidation or setoff of

the collateral in the event of default or failure to re-margin. The effective maturity M_{NS} and floor for such transactions shall be calculated as follows:

(a) For transactions which are not subject to a netting agreement, the effective maturity is calculated as the greater of one day and effective maturity prescribed under sub-paragraph 17(3) above.

(b) For transactions which are subject to a netting agreement, the floor is determined by the minimum holding period for the transaction type, as under:

i. A floor equal to the minimum holding period for the transaction type set out in Table 29 under paragraph 163(8) of the Reserve Bank of India (Commercial Banks - Prudential Norms on Capital Adequacy) Directions, 2025, shall apply to the weighted average maturity of the transactions, as calculated under sub-paragraph 17(4) above.

ii. Where more than one transaction type is contained in the netting agreement, a floor equal to the highest of the minimum holding periods will apply to the average.

Explanation: The collateralisation and documentation conditions specified in the paragraph 17 (5) are specific to the exemption from the one-year maturity floor and do not, by themselves, qualify transactions for treatment as a netting set. Where sub-paragraph 17.5(b) refers to a “netting agreement”, such agreement shall satisfy the applicable requirements for recognition for regulatory capital purposes; in the absence of such recognition, each transaction shall be treated as its own netting set in accordance with sub-paragraph 17.5(a).

B. Full version of the BA-CVA

18. The full version of BA-CVA recognises the effect of counterparty credit spread hedges. The transactions used for the purpose of mitigating the counterparty credit spread component of CVA risk, and managed as such by a bank, shall be eligible hedges.

19. Only single-name credit default swaps (CDS), single-name contingent CDS and index CDS can be eligible CVA hedges. A bank is currently not permitted to engage in single-name contingent CDS. Accordingly, full version of BA-CVA shall

not recognise the effect of counterparty credit spread hedges in single-name contingent CDS.

Provided that, eligible single-name CDS shall meet one of the following criteria:

(i) reference the counterparty directly; or

(ii) reference an entity legally related to the counterparty, where legally related refers to cases where the reference name and the counterparty are either a parent and its subsidiary or two subsidiaries of a common parent; or

(iii) reference an entity that belongs to the same sector and region as the counterparty.

20. A bank that uses the full version of BA-CVA shall also calculate $\mathrm{K}_{\mathrm{reduced}}$ as provided in paragraph 14 of these Directions. The capital requirements for CVA risk under the full version of BA-CVA is calculated using the following formula:

$$CVA_{full} = DS \times K_{full}$$

Where,

DS is the discount scalar = 0.65; and

$\mathrm{K}_{\mathrm{full}}$ is calculated as provided in paragraph 21 of these Directions.

21. A bank shall compute $\mathrm{K}_{\mathrm{full}}$ using the following formula:

$$K_{full} = \beta * K_{reduced} + (1 - \beta) * K_{hedged}$$

Where,

(i) β=0.25, is the supervisory parameter used to provide a floor that limits the extent to which hedging can reduce the capital requirements for CVA risk;

(ii) $\mathrm{K}_{\mathrm{reduced}}$ is calculated as provided in paragraph 14 of these Directions; and

(iii) $\mathrm{K}_{\mathrm{hedged}}$ is the part of the capital requirements that recognises eligible hedges, and is calculated as prescribed in paragraph 22 of these Directions.

22. A bank shall compute $\mathrm{K}_{\mathrm{hedged}}$ using the following formula:

$$K_{hedged} = \sqrt{\left(\rho * \sum_{C}(SCVA_C - SNH_C) - IH\right)^2 + (1 - \rho^2) * \sum_{C}(SCVA_C - SNH_C)^2 + \sum_{C} HMA_C}$$

Where,

(i) the summations shall be taken over all counterparties c that are within the scope of the CVA charge;

(ii) The first term, $(\rho * \sum_{C}(\mathrm{SCVA_C} - \mathrm{SNH_C}) - \mathrm{IH})^2$, aggregates the systematic components of CVA risk arising from the bank's counterparties, single-name hedges and index hedges;

(iii) The second term, $(1 - \rho^2) * \sum_{C}(\mathrm{SCVA_C} - \mathrm{SNH_C})^2$, aggregates the idiosyncratic components of CVA risk arising from the bank's counterparties and single-name hedges;

(iv) The third term $\sum_{C} \mathrm{HMA_C}$ aggregates the components of indirect hedges that are not aligned with counterparties' credit spreads;

(v) ρ is the supervisory correlation parameter, prescribed under paragraph 14(iii) of these Directions;

(vi) $\mathrm{SCVA_C}$ is the stand-alone CVA capital requirement for counterparty c, to be calculated as under paragraph 15 of these Directions;

(vii) SNH_C is a quantity that gives recognition to the reduction in CVA risk of the counterparty c arising from the bank's use of single-name hedges of credit spread risk and shall be calculated as prescribed in paragraph 23 of these Directions;

(viii) IH is a quantity that gives recognition to the reduction in CVA risk across all counterparties arising from the bank's use of index hedges and to be calculated as prescribed in paragraph 24 of these Directions;

(ix) HMA_C is a quantity characterising hedging misalignment, which is designed to limit the extent to which indirect hedges can reduce capital requirements given that they will not fully offset movements in a counterparty's credit spread. In other words, with indirect hedges present, K_{hedged} cannot reach zero and shall be calculated as prescribed in paragraph 25 of these Directions.

23. A bank shall compute $\mathrm{SNH_C}$ using the following formula:

$$SNH_C = \sum_{h \epsilon c} r_{hc} * RW_h * M_h^{SN} * B_h^{SN} * DF_h^{SN}$$

Where,

(i) the summation is across all single name hedges h that the bank has taken out to hedge the CVA risk of counterparty c;

$\mathrm{r_{hc}}$ is the supervisory prescribed correlation between the credit spread of counterparty c and the credit spread of a single-name hedge h of counterparty c. The value of $\mathrm{r_{hc}}$ is set out in the **Table 2** below. The value of $\mathrm{r_{hc}}$ is set at 100 per cent if the hedge directly references the counterparty c and set at lower values if it does not.

(ii) $\mathrm{RW_h}$ is the supervisory risk weight of single-name hedge h that reflects the volatility of the credit spread of the reference name of the hedging instrument. These risk weights are based on a combination of the sector and the credit quality of the reference name of the hedging instrument as prescribed in **Table 1** under paragraph 16 of these Directions.

(iii) $\mathrm{M_h^{SN}}$ is the remaining maturity of single-name hedge h.

(iv) $\mathrm{B_h^{SN}}$ is the notional of single-name hedge h. For single-name contingent CDS, the notional is determined by the current market value of the reference portfolio or instrument. A bank is currently not permitted to engage in single-name contingent CDS.

(v) $\mathrm{DF_h^{SN}}$ is the supervisory discount factor, which shall be calculated using the following formula:

$$DF_h^{SN} = \frac{1 - e^{-0.05 * M_h^{SN}}}{0.05 * M_h^{SN}}$$

Table 2 - Supervisory prescribed correlations between credit spread of counterparty and single-name hedge (r_{hc})

Sl. No.	Type of relation between single-name hedge h of counterparty c	Value of r_{hc}
1	In cases where the single-name hedge h references counterparty c directly	100%

2	In cases where the single-name hedge h has legal relation with counterparty c	80%
3	In cases where the single-name hedge h shares sector and region with counterparty c	50%

24. A bank shall calculate IH using the following formula:

$$IH = \sum_i RW_i * M_i^{ind} * B_i^{ind} * DF_i^{ind}$$

Where,

(i) the summation is across all index hedges i that the bank has taken out to hedge the CVA risk.

(ii) $\mathrm{RW_i}$ is the supervisory risk weight of the index hedge i. It shall be calculated using the risk weights as provided in **Table 1** under paragraph 16 of these Directions, which are based on the sector and the credit quality of the index constituents, and adjusted as follows:

(a) For an index where all index constituents belong to the same sector and are of the same credit quality, the relevant value in the **Table 1** under paragraph 16 of these Directions shall be multiplied by 0.7 to account for diversification of idiosyncratic risk within the index.

(b) For an index spanning multiple sectors or with a mixture of investment grade constituents and other grade constituents, the name-weighted average of the risk weights from the **Table 1** under paragraph 16 of these Directions shall be calculated and then multiplied by 0.7.

(iii) $\mathrm{M_i^{ind}}$ is the remaining maturity of index hedge i.

(iv) $\mathrm{B_i^{ind}}$ is the notional of the index hedge i.

(v) $\mathrm{DF_i^{ind}}$ is the supervisory discount factor, which shall be calculated using the following formula:

$$DF_i^{ind} = \frac{1 - e^{-0.05 * M_i^{ind}}}{0.05 * M_i^{ind}}$$

25. A bank shall calculate $\mathrm{HMA_C}$ using the following formula:

$$HMA_C = \sum_{h \epsilon c} (1 - r_{hc}^2) * (RW_h * M_h^{SN} * B_h^{SN} * DF_h^{SN})^2$$

Where,

(i) the summation is across all single name hedges h that have been taken out to hedge the CVA risk of counterparty c; and

(ii) r_{hc}, RW_h , M_h^{SN}, B_h^{SN}, and DF_h^{SN} have the same definitions as set out in paragraph 23 of these Directions.

Chapter-V: Capital Treatment of CVA Hedges

26. A bank may use CVA hedging instruments that are external (i.e., with an external counterparty) or internal (i.e., with one of the bank's trading desks). Internal CVA involves two perfectly offsetting positions, one of the CVA desk and the opposite position of the trading desk. The capital treatment of CVA hedges is as follows:

(1) Capital treatment of external CVA hedges

(a) A bank shall include in the CVA calculation of the counterparty providing the hedge, all external CVA hedges (including both eligible and ineligible external CVA hedges) that are covered transactions.

(b) A bank shall exclude all eligible external CVA hedges from its market risk capital requirement.

(c) Ineligible external CVA hedges shall be treated as trading book instruments and shall be capitalised for market risk capital requirement.

(2) Capital treatment of internal CVA hedges

(a) If an internal CVA hedge is ineligible, both positions belong to the trading book where they cancel each other. Hence there is no impact on either the CVA portfolio or the trading book.

(b) If an internal CVA hedge is eligible, the CVA desk's position is part of the CVA portfolio where it is capitalised for CVA risk, while the trading desk's position is part of the trading book where it is capitalised for market risk capital requirement.

(3). Eligibility criteria of CVA hedges are specified in paragraphs 18 to 19 of these Directions.

Chapter-VI: Disclosures and Reporting Requirements

27. A bank shall make disclosures on CVA risk, as part of its Pillar 3 disclosure requirements. The disclosure requirements under these Directions are:
    (a) Table CVAA - General qualitative disclosure requirements related to CVA.
    (b) Template CVA1 - The reduced basic approach for CVA (BA-CVA).
    (c) Template CVA2 - The full basic approach for CVA (BA-CVA).
28. The detailed templates and disclosure frequency are provided in **Annex 2**.

Chapter-VII: Repeal

29. Upon coming into effect of these Directions on April 01, 2027, para 85(3) of Reserve Bank of India (Commercial Banks - Prudential Norms on Capital Adequacy) Directions, 2025 shall stand repealed.

(Sunil T S Nair)
Chief General Manager

Annex 1

Illustrative examples of the application of the CVA Framework

Note - The calculations are for illustrative purposes only to aid banks in their understanding of the Directions.

1). **Example 1: Reduced version of the BA-CVA (hedges are not recognised)**

The table below summarises the transactions that are within the scope of the CVA charge. For the sake of simplicity, this example assumes that the application of SA-CCR results in EAD for the respective counterparties mentioned in the table. The values obtained in calculations for the sample portfolios have been rounded to two decimal points for ease of presentation (except for DF values). All values in the table are given in INR crore.

Counterparty	**Counterparty 1**	**Counterparty 2**
Credit quality	Investment grade	Investment grade
Sector of counterparty	Bank	Corporate-Industrial
Risk weight	5.00%	3.00%
Effective Maturity	3 years	2 years
EAD	₹1000	₹600

Step 1: Calculate the $\mathrm{DF_{NS}}$ using the following formula

$$DF_{NS} = \frac{1 - e^{-0.05 * M_{NS}}}{0.05 * M_{NS}}$$

DF is the supervisory discount factor averaged over time between today and the netting set's effective maturity date. The interest rate used for discounting is set at 5 per cent, hence 0.05 in the formula.

Counterparty 1	**Counterparty 2**
$DF_{NS} = \frac{1 - e^{-0.05 * M_{NS}}}{0.05 * M_{NS}}$ $DF_{NS} = \frac{1 - e^{-0.05 * 3}}{0.05 * 3}$ $DF_{NS} = 0.929$	$DF_{NS} = \frac{1 - e^{-0.05 * M_{NS}}}{0.05 * M_{NS}}$ $DF_{NS} = \frac{1 - e^{-0.05 * 2}}{0.05 * 2}$ $DF_{NS} = 0.952$

Step 2: Calculate the stand-alone CVA capital requirements for counterparty using the following formula

$$SCVA_C = \frac{1}{\alpha} * RW_C * \sum_{NS} M_{NS} * EAD_{NS} * DF_{NS}$$

Counterparty 1	**Counterparty 2**
α = 1.4	α = 1.4
M_{NS} = 3	$M_{NS} = 2$
RW_C = 5%	RW_C = 3%
EAD = 1000	EAD = 600
DF_{NS} = 0.929	$DF_{NS} = 0.952$
$SCVA_{C1} = \frac{1}{1.4} * 0.05 * (3 * 1000 * 0.929)$ = 99.49	$SCVA_{C2} = \frac{1}{1.4} * 0.03 * (2 * 600 * 0.952)$ =24.47

Step 3: Calculate the portfolio CVA

Computation of Portfolio CVA

$\sum_C SCVA_C = 99.49 + 24.47 = 123.96$

$\sum_C SCVA_C{}^2 = 99.49\text{^}2 + 24.47\text{^}2 = 9899.12 + 598.80 = 10497.92$

$\rho = 0.50$

$(1 - \rho^2) = 0.75$

Step 4: Calculate the $K_{reduced}$

$$K_{reduced} = \sqrt{\left(\rho * \sum_C SCVA_C\right)^2 + (1 - \rho^2) * \sum_C SCVA_C{}^2}$$

$$K_{reduced} = \sqrt{(0.5 * 123.96)^2 + 0.75 * 10497.92}$$

$$K_{reduced} = \sqrt{3842 + 7873}$$

$$K_{reduced} = 108.24$$

Step 5: Calculate the capital requirements for CVA risk under the reduced version of the BA-CVA using the following formula

Application of Discount Scalar

$$CVA_{reduced} = DS \times K_{reduced}$$

=0.65*108.24

= 70.35

2). **Example 2: Full version of the BA-CVA**

The table below summarises the transactions that are within the scope of the CVA charge. All values in the table are given in INR crore. For the sake of simplicity, this example assumes that the application of SA-CCR results in EAD for the respective counterparties mentioned in the table. The values obtained in calculations for the sample portfolios have been rounded to two decimal points for ease of presentation (except for DF values). All values in the table are given in INR crore.

Counterparty	**Counterparty 1**	**Counterparty 2**
Credit quality	Investment grade	Investment grade
Sector of counterparty	Bank	Corporate-Industrial
Risk weight (RW_h) (RW_C)	5.00%	3.00%
Effective Maturity $(\mathrm{M_{NS}})$	3 years	2 years
EAD	₹1000	₹600
Hedging Instrument	Single name hedge on Counterparty 1	Single name hedge on Counterparty 2
Notional of the hedge	₹400	₹200
Effective Maturity of the hedge (M_h^{SN})	3 years	2 years

Step 1: Calculate the $\mathrm{DF_h^{SN}}$ using the following formula

$$DF_h^{SN} = \frac{1 - e^{-0.05 * M_h^{SN}}}{0.05 * M_h^{SN}}$$

DF is the supervisory discount factor averaged over time between today and the netting set's effective maturity date. The interest rate used for discounting is set at 5 per cent, hence 0.05 in the formula.

Counterparty 1	Counterparty 2
$DF_h^{SN} = \frac{1 - e^{-0.05 * M_h^{SN}}}{0.05 * M_h^{SN}}$ $DF_h^{SN} = \frac{1 - e^{-0.05 * 3}}{0.05 * 3}$ $DF_h^{SN} = 0.929$	$DF_h^{SN} = \frac{1 - e^{-0.05 * M_h^{SN}}}{0.05 * M_h^{SN}}$ $DF_h^{SN} = \frac{1 - e^{-0.05 * 2}}{0.05 * 2}$ $DF_h^{SN} = 0.952$

Step 2: Calculate $\mathrm{SNH_C}$ using the following formula:

$$SNH_C = \sum_{h \epsilon c} r_{hc} * RW_h * M_h^{SN} * B_h^{SN} * DF_h^{SN}$$

The summation is across all single name hedges h that the bank has taken out to hedge the CVA risk of counterparty c.

Counterparty 1	Counterparty 2
$r_{hc} = 100\%$	$r_{hc} = 100\%$
RW_h =5%	$RW_h = 3\%$
$M_h^{SN} = 3\ years$	$M_h^{SN} = 2\ years$
$\mathrm{B_h^{SN}} =$ ₹400 $crore$	$\mathrm{B_h^{SN}} =$ ₹200 $crore$
$DF_h^{SN} = 0.929$	$DF_h^{SN} = 0.952$
$SNH_{C1} = 1 * 0.05 * 3 * 400 * .929$ $SNH_{C1} = 55.74$	$SNH_{C2} = 1$ *0.03*2*200*.952 $SNH_{C2} = 11.42$

Step 3: Calculate $\mathrm{HMA_C}$ using the following formula:

$$HMA_C = \sum_{h \epsilon c} (1 - r_{hc}^2) * (RW_h * M_h^{SN} * B_h^{SN} * DF_h^{SN})^2$$

Where,

the summation is across all single name hedges h that have been taken out to hedge the CVA risk of counterparty c;

Counterparty 1	Counterparty 2
$r_{hc} = 100\%$	$r_{hc} = 100\%$
RW_h =5%	$RW_h = 3\%$
$M_h^{SN} = 3\ years$	$M_h^{SN} = 2\ years$
$\mathrm{B_h^{SN}} =$ ₹$400\ crore$	$\mathrm{B_h^{SN}} =$ ₹$200\ crore$
$DF_h^{SN} = 0.929$	$DF_h^{SN} = 0.952$
$HMA_{C1} = (1\text{-}1^2) * (0.05 * 3 * 400 * 0.929)^2$ $= 0$	$HMA_{C2} = (1\text{-}1^2) * (0.03 * 2 * 200 * 0.952)^2$ $= 0$

Step 4: Calculate the stand-alone CVA capital requirements for counterparty using the following formula

$$SCVA_C = \frac{1}{\alpha} * RW_C * \sum_{NS} M_{NS} * EAD_{NS} * DF_{NS}$$

$SCVA_{C1}\ is\ 99.49\ and\ SCVA_{C2}\ is\ 24.47\ as\ derived\ in\ Example\ 1.$

Step 5: Calculate $SCVA_C$ - SNH_C using the following formula:

Counterparty	$SCVA_C$	SNH_C	$SCVA_C$ - SNH_C
Counterparty 1	99.49	55.74	43.75
Counterparty 2	24.47	11.42	13.05

Step 6: Calculate K_{hedged} using the following formula:

$$K_{hedged} = \sqrt{\left(\rho * \sum_C (SCVA_C - SNH_C) - IH\right)^2 + (1 - \rho^2) * \sum_C (SCVA_C - SNH_C)^2 + \sum_C HMA_C}$$

$\sum_C (SCVA_C - SNH_C)$	43.75+13.05=56.80
IH	0 (Index hedge is not used in this illustration)
ρ	0.50
$(1 - \rho^2)$	0.75

$\sum_C (SCVA_C - SNH_C)^2$	43.75^2 + 13.05^2 = 1914.06 + 170.30 = 2084.36
$\sum_C HMA_C$	0

$$K_{hedged} = \sqrt{(0.50 * (56.80 - 0))^2 + 0.75 * 2084.36}$$

$$K_{hedged} = \sqrt{806.56 + 1563.27}$$

$$K_{hedged} = \sqrt{2369.83}$$

$$K_{hedged} = 48.68$$

Step 7: Calculate $K_{reduced}$ using the following formula:

$$K_{reduced} = \sqrt{\left(\rho * \sum_C SCVA_C\right)^2 + (1 - \rho^2) * \sum_C {SCVA_C}^2}$$

$K_{reduced}$ = 108.24 *(As calculated in example 1)*

Step 8: Calculate K_{full} using the following formula:

$$K_{full} = \beta * K_{reduced} + (1 - \beta) * K_{hedged}$$

β	0.25
$K_{reduced}$	108.24
$(1 - \beta)$	0.75
K_{hedged}	48.68
K_{full}	63.57

Step 8: Calculate the capital requirements for CVA risk under the reduced version of the BA-CVA using the following formula

$$CVA_{full} = DS \times K_{full}$$

$$CVA_{full} = 0.65 * 63.57$$

$$CVA_{full} = 41.32$$

Annex 2

Pillar 3 Disclosure Tables and Templates on CVA risk

1. Table CVAA - General qualitative disclosure requirements related to CVA

Purpose: To provide a description of the risk management objectives and policies for CVA risk.	
Scope of application: The table is mandatory for a bank that is subject to CVA capital requirements, including a bank which is qualified and has elected to set its capital requirement for CVA at 100 per cent of its counterparty credit risk charge as per the Alternate Treatment in accordance with paragraphs 8 and 9 of these Directions.	
Content: Qualitative information.	
Frequency: Annual.	
Format: Flexible.	
A bank shall describe its risk management objectives and policies for CVA risk as follows:	
(a)	An explanation and / or a description of the bank's processes implemented to identify, measure, monitor and control the bank's CVA risks, including policies for hedging CVA risk and the processes for monitoring the continuing effectiveness of hedges.
(b)	Whether the bank is eligible and has chosen to set its capital requirement for CVA at 100 per cent of the bank's capital requirement for counterparty credit risk as per paragraphs 8 and 9 of these Directions.

2. Template CVA1: The reduced basic approach for CVA (BA-CVA)

Purpose: To provide the components used for the computation of RWA under the reduced version of BA-CVA for CVA risk.			
Scope of application: The template is mandatory for a bank having its RWA for CVA risk measured according to the reduced version of BA-CVA.			
Content: RWA.			
Frequency: Semiannual.			
Format: Fixed.			
Accompanying narrative: A bank shall describe the types of hedges it uses even if they are not taken into account under the reduced BA-CVA.			
		(a)	(b)
		Components	BA-CVA RWA
1	Aggregation of systematic components of CVA risk		
2	Aggregation of idiosyncratic components of CVA risk		
3	Total		

Definitions and instructions

Row number	Explanation
1	Aggregation of systematic components of CVA risk: RWA under perfect correlation assumption $(\sum_C SCVA_C)$ as per paragraph 14 of these Directions.
2	Aggregation of idiosyncratic components of CVA risk: RWA under zero correlation assumption $\sqrt{(\sum_C SCVA_C{}^2)}$ as per paragraph 14 of these Directions. *Note:* This term is the square root of the sum term $\sum_C SCVA_C{}^2$ term used in the formula for $K_{reduced}$
3	Total: $CVA_{reduced}$ as per paragraph 13 of these Directions, multiplied by 12.5.

3. Template CVA2: The full basic approach for CVA (BA-CVA)

Purpose: To provide the components used for the computation of RWA under the full version of BA-CVA for CVA risk.		
Scope of application: The template is mandatory for a bank having its RWA for CVA risk measured according to the full version of BA-CVA.		
Content: RWA.		
Frequency: Semiannual.		
Format: Fixed. Additional rows can be inserted for the breakdown of other risks.		
		(a)
		BA-CVA RWA
1	K reduced	
2	K hedged	
3	Total	

Definitions and instructions

Row number	Explanation
1	K reduced: $K_{reduced}$ as per paragraph 14 of these Directions.
2	K hedged: K_{hedged} as per paragraph 22 of these Directions.
3	Total: CVA_{full} as per paragraph 20 of these Directions, multiplied by 12.5.